

Course Project

Using AI to Write Code and Train Neural Networks

使用AI编写代码并训练神经网络

任务2.分割

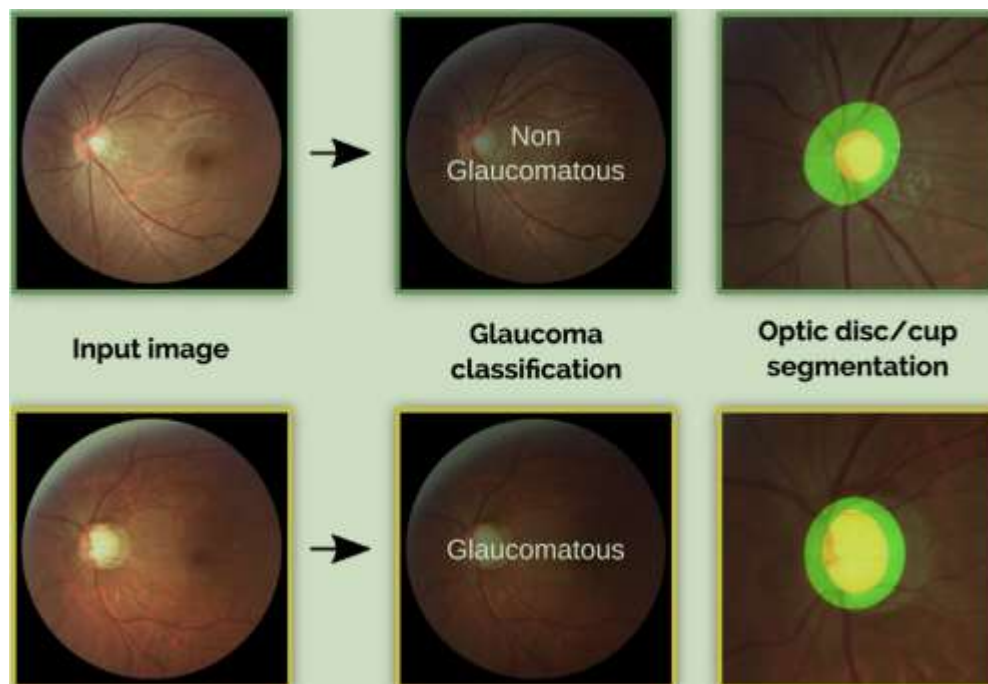
Task2.Segmentation

数据集: REFUGE

Dataset: REFUGE

下载链接(Download Link)

https://aistudio.baidu.com/datasetdetail/320358?login_type=qzone



Optic disc/cup segmentation
视杯和视盘分割

Task2.1 认识数据集

Task2.1 Understanding the Dataset

- 1.成功加载 REFUGE 数据集，解决环境配置与路径问题。
- 2.展示图像image和他们的标签label
- 3.简单描述在图中什么区域是视盘、什么是视杯，他们在图中有什么显而易见的特征

1. Successfully loaded the REFUGE dataset and resolved environment configuration and path issues.
2. Displayed the images and their corresponding labels.
3. Briefly described which areas in the images correspond to the optic disc and which to the optic cup, and highlighted their most obvious characteristics in the images.

Task2.2 网络调研和选择

Task2.2 Network Research and Selection

- 1.了解当前主流的图像分割网络架构，并简单介绍各个类型的网络
- 2.自主挑选一个网络架构作为你本次大作业的基座模型并给出理由

1. Understand the current mainstream image segmentation network architectures and briefly introduce each type of network.
2. Independently select one network architecture as the base model for your major assignment and provide a rationale for your choice.

Task2.3 AI辅助编写你挑选网络架构的代码

Task2.3 AI-assisted code writing for selecting a network architecture

- 1.对比分类任务，哪些是分割任务特有的，比如特殊的损失函数，Logits输出的维度变化等
- 2.了解不同损失函数带来的不同优化方向 (eg:DICE, Cross-Entropy Loss...)
- 3.了解如何评估分割模型的性能 (eg:IoU、DICE...)
- 4.实验对比不同损失函数(DICE, Cross-Entropy Loss分别使用)的效果，以及组合损失函数 (DICE+Cross-Entropy Loss) ,记录Loss曲线变化，评估你的分割模型性能，并分析原因

1. Compare and contrast segmentation tasks with classification tasks, identifying which aspects are unique to segmentation tasks—for example, specialized loss functions and changes in the dimensionality of logits output.
2. Understand how different loss functions lead to distinct optimization directions (e.g., DICE, Cross-Entropy Loss).
3. Learn how to evaluate the performance of segmentation models (e.g., IoU, DICE).
4. Conduct experiments to compare the performance of segmentation models using different loss functions (DICE and Cross-Entropy Loss separately) as well as a combined loss function (DICE + Cross-Entropy Loss). Record the changes in the loss curves, evaluate your segmentation model's performance, and analyze the underlying reasons for these results.

Task2.4 模型参数调整与优化

Task2.4 Model Parameter Tuning and Optimization

1. 列出你所选模型中的哪些参数是可调的、并简单介绍 (eg: 学习率 Learning Rate、批次大小 Batch Size、学习率调度器 Scheduler...)
2. **实验对比** 不同的参数对模型的影响, 观察记录训练时长、Loss 曲线等变化 (收敛速度)
3. 分析不同参数配造成你所记录的现象的原因

1. List which parameters in your selected model are adjustable, and briefly describe them (e.g., Learning Rate, Batch Size, Learning Rate Scheduler...).
2. Conduct experiments to compare the impact of different parameters on the model, and observe and record changes such as training duration and loss curves (including convergence speed).
3. Analyze the reasons behind the phenomena you've observed under different parameter configurations.

2.5 Optional Task

2.5.1分割效果的评估和后处理

- 1.孔洞检查，在部分分割结果中，由于分割效果不佳会出现一个孔洞，正常眼是封闭的完整的
- 2.连通域检查，一个正常的眼，只会有一个视盘和视杯，也就是两个连通域，出现多的连通域是什么情况？
- 3.设计算法修复孔洞和连通域，并给出设计思路和修复前修复后的对比

2.5.1Evaluation and Post-processing of Segmentation Results

- 1.Hole Checking: In some segmentation results, a hole may appear due to poor segmentation performance. A normal eye is closed and complete.
- 2.Connected Component Checking: A normal eye will only have one optic disc and one optic cup, which means two connected components. What does it mean if more connected components appear?
- 3.Design an algorithm to repair holes and connected components, and provide the design idea along with a comparison before and after repair.

2.5 Optional Task

2.5.2 拓扑结构约束

- 1.检查分割结果，在视杯分割中，视杯必须包含在视盘内部（拓扑约束），是否出现了视杯超出了视盘边界。
- 2.自定义“带约束”的损失函数，并讲解设计思路和理由。（损失函数能否梯度回传，能否反向传播）
- 3.展示“带约束”的损失函数和普通损失函数在“难样本”样本上的分割表现。

2.5.2 Topological Structure Constraints

- 1.Check the segmentation results. In optic cup segmentation, the optic cup must be contained within the optic disc (topological constraint). Check if the optic cup has exceeded the boundary of the optic disc.
- 2.Customize a "constrained" loss function and explain the design idea and reasons. (Can the loss function support gradient backpropagation? Can it backward?)
- 3.Demonstrate the segmentation performance of the "constrained" loss function versus the standard loss function on "hard samples".

